Option Pricing under Stochastic Volatility with Variance Reduction via Rasmussen Control Variates

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"*For the things we have to learn before we can do them,*
we learn by doing them."

Aristotle

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Abstract

American options are widely traded in financial markets yet their valuation remains a complex problem that has been the subject of extensive academic research. The least squares Monte Carlo (LSM) approach proposed by Longstaff and Schwartz (2001) continues to stand as one of the most competitive methods, particularly when it can be combined with variance reduction techniques. For example, under the Black-Scholes model, the use of control variates based on European options as proposed by Rasmussen (2002) yields significant improvements in accuracy of LSM, outperforming machine learning methods like neural nets as reported in Aquino et al. (2022). This approach however, presupposes a closed formula for the European option price available for models such as Black and Scholes (1973), Stein and Stein (1991), and Schöbel and Zhu (1999). More recently, Lin et al. (2024) proposed a stochastic volatility model that nests the aforementioned models and they derived a semi-analytical approach for pricing European options following the techniques developed by Heston (2015). The authors use Fourier inversion and ODE numerics to estimate the European price and they report to yield the result more efficiently than with Monte Carlo approach. This efficiency suggests their formula as an attractive candidate for computing the European price in Rasmussen's framework. In the present work, we conduct European pricing experiments from Lin et al. (2024) and we estimate the American put price under their stochastic volatility setting. The price estimation combines least squares Monte Carlo and Rasmussen control variates following Rasmussen (2005) and West (2013). We demonstrate this valuation approach to remain computationally feasible and highly effective for variance reduction despite computing European prices with a semi-analytical formula. Experiments show that accuracy and stability of the price estimator can be attained at lower count of Monte Carlo paths, in agreement with the results in Rasmussen (2005).

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1 Introduction

This document presents a numerical study of option pricing under the stochastic volatility model (1)-(3) proposed by Lin et al. (2024), hereafter referred to as the LLH model. The following sections present the model and provide the background for each pricing approach used and report on the conducted experiments.

The LLH model is presented in section 2 together with discretization schemes and an algorithm for Monte Carlo simulations. The pricing methods for European calls and American puts are developed in section 3. A summary of the semi-analytical European pricing approach derived in Lin et al. (2024) is provided followed by background on the valuation of American options using least-squares Monte Carlo (LSM) Longstaff and Schwartz (2001), Aquino et al. (2022). A brief on Rasmussen's control variate framework is given following Rasmussen (2002) and West (2013).

Experiments and findings are reported in section 4. European and American vanilla options are priced across moneyness levels and parameter regimes. The implementation of the semi-analytical European price is tested against independently evaluated prices from the edge-case models of Black–Scholes (BS), Stein–Stein (SS), and Schöbel–Zhu (SZ). Several LSM price estimators, plain and with control variate under varying exercise floors are considered. Variance reduction and early exercise premium (EEP) are computed and reported. A summary of the findings and conclusions from the conducted experiments is presented in section 5. The report concludes with notes on improvement venues and future work.

2 The LLH Model

The LLH model proposed in Lin et al. (2024) extends the Schöbel and Zhu (1999) model by allowing a time-varying long-run mean volatility θ_t, driven by a standard Brownian motion and constant drift. The SZ model improves upon the Stein and Stein (1991) model by allowing correlated Brownian motions that govern the price S_t and its volatility σ_t. The SS model improves on Black and Scholes (1973) model by allowing an Ornstein-Uhlenbeck stochastic volatility σ_t while imposing uncorrelated Brownian drivers.

2.1 Mathematical Formulation

The LLH model specifies the following system of stochastic differential equations under the risk-neutral measure:

$$\frac{dS_t}{S_t} = r\,dt + \sigma_t\,dW_t^1, \tag{1}$$

$$d\sigma_t = \kappa(\theta_t - \sigma_t)\,dt + \nu\,dW_t^2 \tag{2}$$

$$d\theta_t = \lambda\,dt + \eta\,dW_t \tag{3}$$

where W_t^1, W_t^2, and W_t are Brownian motions with $\langle W^1, W^2\rangle_t = \rho t$ and W_t independent of W_t^1 and W_t^2. Here, r is the risk-free interest rate, κ is the reversion rate of the volatility σ_t to its long-term mean θ_t, ν is the volatility of σ_t (vol-of-vol parameter), λ and η are the drift and volatility of θ_t, and ρ is the correlation coefficient of the Brownian drivers of the price and volatility dynamics. The unobserved parameter set is $\Theta = (\kappa, \nu, \lambda, \eta, \rho, \sigma_0, \theta_0)$. See Table 1 for the parameter values used throughout this work and Figure 1 for a plot of the model dynamics under the considered parameter regimes.

Remark 2.1 LHH model is reduced to the special-case nested models by setting certain parameters to 0: BS model ($\kappa = \nu = \lambda = \eta = \rho = 0$), SS model ($\lambda = \eta = \rho = 0$), and SZ model ($\lambda = \eta = 0$).

Remark 2.2 Note that if $x_t = \ln S_t$ is introduced, then Itô's lemma applied to x_t and direct integration of equation (3) allow the SDE system (1)-(3) to be reduced to:

$$dx_t = \left(r - \tfrac{1}{2}\sigma_t^2\right) dt + \sigma_t \, dW_t^1, \tag{4}$$
$$d\sigma_t = \kappa(\theta_t - \sigma_t) \, dt + \nu \, dW_t^2 \tag{5}$$
$$d\theta_t = \lambda \, dt + \eta \, dW_t. \tag{6}$$

It follows from (4)-(6) that the dynamics described by the LLH are that of the vector Itô process $\boldsymbol{X}_t = \begin{bmatrix} x_t \\ \sigma_t \\ \theta_t \end{bmatrix}$ governed by $d\boldsymbol{X}_t = \boldsymbol{a}(t, \boldsymbol{X}_t)dt + b_1(t, \boldsymbol{X}_t)d\boldsymbol{W}_t$, with drift vector and diffusion matrix:

$$\boldsymbol{a}(t, \boldsymbol{X}_t) = \begin{bmatrix} r - \frac{1}{2}\sigma_t^2 \\ \kappa(\theta_t - \sigma_t) \\ \lambda \end{bmatrix}, \qquad b_1(t, \boldsymbol{X}_t) = \begin{bmatrix} \sigma_t & 0 & 0 \\ 0 & \nu\sqrt{1-\rho^2} & 0 \\ 0 & 0 & \eta \end{bmatrix}, \tag{7}$$

and Brownian driver $d\boldsymbol{W}_t = \begin{bmatrix} dW_t^1 \\ dW_t^2 \\ dW_t \end{bmatrix}$.

Remark 2.3 For simulation purposes, it is useful to rewrite equations (4)-(6) in terms of the independent Brownian motions (Z_t^1, Z_t^2, Z_t^3) defined by:

$$W_t^1 = Z_t^1, \qquad W_t^2 = \rho \, Z_t^1 + \sqrt{1-\rho^2} \, Z_t^2, \qquad W_t = Z_t^3 \tag{8}$$

where the decomposition of W_t^2 is standard for Brownian motions W_t^1 and W_t^2 with correlation ρ. Equations (4)-(6) may now be rewritten as

$$dx_t = \left(r - \tfrac{1}{2}\sigma_t^2\right)dt + \sigma_t \, dZ_t^1 \tag{9}$$
$$d\sigma_t = \kappa(\theta_t - \sigma_t) \, dt + \nu\rho \, dZ_t^1 + \nu\sqrt{1-\rho^2} \, dZ_t^2 \tag{10}$$
$$d\theta_t = \lambda \, dt + \eta \, dZ_t^3 \tag{11}$$

and the Itô process $\boldsymbol{X}_t$ now satisfies the SDE $d\boldsymbol{X}_t = \boldsymbol{a}(t, \boldsymbol{X}_t)dt + b_2(t, \boldsymbol{X}_t)d\boldsymbol{Z}_t$ where

$$b_2(t, \boldsymbol{X}_t) = \begin{bmatrix} \sigma_t & 0 & 0 \\ \nu\rho & \nu\sqrt{1-\rho^2} & 0 \\ 0 & 0 & \eta \end{bmatrix}, \qquad \boldsymbol{Z}_t = \begin{bmatrix} Z_t^1 \\ Z_t^2 \\ Z_t^3 \end{bmatrix} \tag{12}$$

2.2 Simulation from the LLH Model

We suggest two schemes for the discretization of the LLH model. An Euler scheme as a benchmark and a Milstein scheme derived for equations (9)-(11) following section 6.2 in Glasserman (2004). The former method is weak order 1 and the latter is weak order 2.

2.2.1 Euler's Scheme

This scheme is based on the discretization $\hat{\boldsymbol{X}}_{n+1} = \hat{\boldsymbol{X}}_n + \boldsymbol{a}(t_n, \hat{\boldsymbol{X}}_n)\Delta t + b_1(t_n, \hat{\boldsymbol{X}}_n)\,\Delta\boldsymbol{W}_n$ which implies the following recursion equations for LLH dynamics (1)-(3):

$$\hat{S}_{n+1} = \hat{S}_n \exp\left\{\left(r - \frac{1}{2}\sigma_t^2\right)\Delta t + \hat{\sigma}_n \Delta W_n^1\right\} \tag{13}$$
$$\hat{\sigma}_{n+1} = \hat{\sigma}_n + \kappa(\hat{\theta}_n - \hat{\sigma}_n)\Delta t + \nu\Delta W_n^2 \tag{14}$$
$$\hat{\theta}_{n+1} = \hat{\theta}_n + \lambda\Delta t + \eta\Delta W_n \tag{15}$$

2.2.2 Milstein Scheme

For a system of SDEs $dX_t^i = a_i(t, \boldsymbol{X}_t)\,dt + \sum_{k=1}^m b_{ik}(t, \boldsymbol{X}_t)\,dZ_t^k$ driven by independent Brownian motions Z_t^k, a Milstein discretization is given by:

$$\begin{aligned}\hat{X}_{n+1}^i = \hat{X}_n^i + a_i\Delta t + \sum_k b_{ik}\,\Delta Z_n^i + \tfrac{1}{2}\,\mathcal{L}^0 a_i\,\Delta t^2 + \tfrac{1}{2}\sum_k \big(\mathcal{L}^k a_i + \mathcal{L}^0 b_{ik}\big)\,\Delta Z_n^i h \\ + \tfrac{1}{2}\sum_k\sum_j \mathcal{L}^j b_{ik}\,\big(\Delta Z_n^j\,\Delta Z_n^k - V_{jk}\big),\end{aligned} \tag{16}$$

where the differential operators $\mathcal{L}^k$ are as defined in section 6.2, Glasserman (2004). The V_{jk} are independent Bernoulli random variables that are independent of the Brownian increments with

$$\Pr\big(V_{jk} = +h\big) = \Pr\big(V_{jk} = -h\big) = \tfrac{1}{2}, \qquad V_{kj} = -V_{jk}, \qquad V_{jj} := h \tag{17}$$

For the SDE equations (9)-(11), these operators can be shown to have the following expressions:

$$\begin{aligned}\mathcal{L}^0 &= \frac{\partial}{\partial t} + \big(r - \tfrac{1}{2}\sigma_t^2\big)\partial_1 + \kappa(\theta_t - \sigma_t)\partial_2 + \lambda\,\partial_3 \\ &\quad + \tfrac{1}{2}\sigma_t^2\,\partial_{11}^2 + \tfrac{1}{2}\nu^2\,\partial_{22}^2 + \tfrac{1}{2}\,\eta^2\,\partial_{33}^2 + \sigma_t\nu\rho\,\partial_{12}^2, &(18)\\ \mathcal{L}^1 &= \sigma_t\,\partial_1 + \nu\rho\,\partial_2, &(19)\\ \mathcal{L}^2 &= \nu\sqrt{1-\rho^2}\,\partial_2, &(20)\\ \mathcal{L}^3 &= \eta\,\partial_3. &(21)\end{aligned}$$

where $\partial_1, \partial_2, \partial_3$ are partial derivatives with respect to x_t, σ_t, θ_t respectively.

The Milstein scheme for the LLH dynamics is obtained from (16) by applying the operators (18)-(21) to the coefficients of the drift vector (7) and diffusion matrix (12). Reverting to the original Brownian drivers (8) yields the recursion equations:

$$\begin{aligned}\ln\hat{S}_{n+1} &= \ln\hat{S}_n + \big(r - \tfrac{1}{2}\hat{\sigma}_n^2\big)\Delta t + \hat{\sigma}_n\,\Delta W_n^1 \\ &\quad - \big[\kappa\hat{\sigma}_n(\hat{\theta}_n - \hat{\sigma}_n) + \tfrac{1}{2}\nu^2\big]\,\Delta t^2 \\ &\quad + \tfrac{1}{2}\kappa(\hat{\theta}_n - \hat{\sigma}_n)\,\Delta W_n^1\Delta t - \tfrac{1}{2}\nu\hat{\sigma}_n\,\Delta W_n^2\Delta t \\ &\quad + \tfrac{1}{2}\nu\big(\Delta W_n^1\Delta W_n^2 - \rho\,\Delta t - \sqrt{1-\rho^2}\,\xi_n\Delta t\big). &(22)\\ \hat{\sigma}_{n+1} &= \hat{\sigma}_n + \kappa(\hat{\theta}_n - \hat{\sigma}_n)\Delta t + \nu\,\Delta W_n^2 \\ &\quad + \kappa\big[\lambda - \kappa(\hat{\theta}_n - \hat{\sigma}_n)\big]\Delta t^2 \\ &\quad - \tfrac{1}{2}\kappa\nu\,\Delta W_n^2\,\Delta t + \tfrac{1}{2}\kappa\,\eta\,\Delta W_n\,\Delta t, &(23)\\ \hat{\theta}_{n+1} &= \hat{\theta}_n + \lambda\Delta t + \eta\,\Delta W_n, &(24)\end{aligned}$$

where ξ_n is a fair Bernoulli random variable taking values ± 1.

2.3 Simulation Parameters

The reference parameter values used across all performed experiments and shown in Table 1 are borrowed from Lin et al. (2024). These parameter sets present two qualitatively different regimes of the model (1)-(3). LLH Table 1 generates more realistic dynamics than LLH Table 2 which introduces negative values of σ_t and θ_t, see Figure 1. We use the former parameter set to validate and interpret our option pricing experiments, while the latter is only used to validate the pricing results against anomalous values of the state variables.

Table 1: Reference parameter sets

	r	κ	ν	λ	η	ρ	σ_0	θ_0
LLH Table 1	0.01	5.0	0.2	0.9	0.01	-0.2	0.15	0.18
LLH Table 2	0.01	4.9394	0.3943	0.3115	0.4112	0.1691	0.2924	0.1319

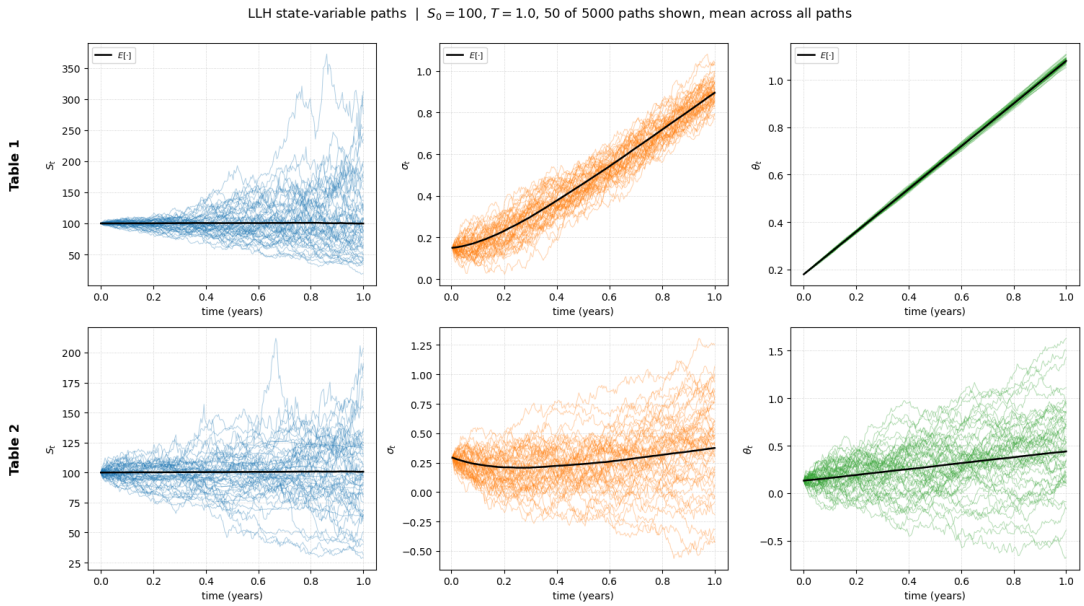


Figure 1: Simulated paths of S_t, σ_t, and θ_t — $\tau = 1.0$, 50 paths.

3 Pricing Approach

In this section, we describe the pricing approaches of European and American options that have been applied in the conducted experiments.

3.1 European Call Price

Let U denote the time t price of a European call with spot S_t, strike K, maturity date T, and time to maturity $\tau = T - t$.

3.1.1 Monte Carlo Price

Under risk-neutral measure $\mathbb{Q}$, the European option price is given by

$$U(\tau, S_t, \sigma_t, \theta_t, K) = \mathbb{E}_t^{\mathbb{Q}} \left[e^{-r\tau} \left(S_T - K \right)^+ \right] \tag{25}$$

Thus, $U(\tau, S_t, \sigma_t, \theta_t, K)$ may be estimated using a Monte Carlo simulation. We apply the following algorithm for computing an estimate $\hat{U}$ of U. Set M to the number of steps from t to T, N the number of paths to simulate, and fix a discretization scheme of the LLH model (4)-(6) (either Euler scheme (13)-(15) or Milstein scheme (22)-(24)).

We simulate the N paths as follows:

1. For $i = 1, 2, 3$, for $k = 1, \ldots, N$, generate independent normal increments ΔZ_k^i. If Milstein scheme is used, also generate Bernoulli values ξ_k taking values ± 1 each with probability $\frac{1}{2}$.

2. Revert to the Brownian incrments ΔW_k^i using (8).

3. For $n = 0, \ldots, M-1$, update x_n, σ_n, θ_n using the fixed scheme.

4. Compute the Monte Carlo price as

$$\hat{U} = \frac{1}{N} \sum_{i=1}^{N} e^{-r\tau} \left(\hat{S}_T - K \right)^+ \tag{26}$$

3.1.2 LLH Price

We summarize the derivation of the pricing approach developed in Lin et al. (2024). The time t price of a European call with strike K and maturity T is computed using

$$U(\tau, S_t, \sigma_t, \theta_t, K) = S_t \, P_1(\tau; K) - K \, e^{-r\tau} \, P_2(\tau; K) \tag{27}$$

where $\tau = T - t$, $P_1 = \mathbb{Q}^\sharp[S_T > K \,|S_t]$, and $P_2 = \mathbb{Q}[S_T > K \,|S_t]$. Here, $\mathbb{Q}$ is the risk-neutral measure and $\mathbb{Q}^\sharp$ has Radon-Nikodym derivative

$$\frac{d\mathbb{Q}^\sharp}{d\mathbb{Q}} = e^{-r(T-t)} \frac{S(T)}{S(t)}. \tag{28}$$

The probabilities P_j can be recovered from the characteristic functions f_j via the Fourier inversion:

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \mathrm{Re}\left[\frac{e^{-i\varphi \ln K}}{i\varphi} \, f_j(\tau; \varphi) \right] d\varphi, \quad j = 1, 2 \tag{29}$$

where $f_1(\phi) = \mathbb{E}_t^{\mathbb{Q}^\sharp}[e^{i\phi x_T}]$, and $f_2(\phi) = \mathbb{E}_t^{\mathbb{Q}}[e^{i\phi x_T}]$ are characteristic functions of $x_T = \ln S_T$ conditional on the information available at time t. See Lord and Kahl (2006) and the references therein for more background on (27) and (29).

The backbone of the pricing approach in Lin et al. (2024) is the computation of the characteristic functions using the following semi-analytical approach:

$$f_j(\tau; \phi) = \exp\Big(C + D \, \sigma_t^2 + E \, \sigma_t \theta_t + F \, \theta_t^2 + G \, \theta_t + H \, \sigma_t + i\phi \, x_t \Big), \tag{30}$$

where the coefficients $C(\tau, \phi), D(\tau, \phi), E(\tau, \phi), F(\tau, \phi), G(\tau, \phi), H(\tau, \phi)$ satisfy the autonomous system of ODEs (31), integrated from $\tau = 0$ to the desired maturity:

$$\begin{aligned}
\frac{\partial D}{\partial \tau}(\tau; \phi) &= \left(u_j \, i\phi - \tfrac{1}{2}\phi^2 \right) + 2\nu^2 D^2(\tau; \phi) - 2A_j(\phi) \, D(\tau; \phi) + \tfrac{1}{2}\eta^2 E^2(\tau; \phi), \\
\frac{\partial E}{\partial \tau}(\tau; \phi) &= 2\kappa \, D(\tau; \phi) - A_j(\phi) \, E(\tau; \phi) + 2\nu^2 D(\tau; \phi) E(\tau; \phi) + 2\eta^2 E(\tau; \phi) F(\tau; \phi), \\
\frac{\partial F}{\partial \tau}(\tau; \phi) &= \kappa \, E(\tau; \phi) + \tfrac{1}{2}\nu^2 E^2(\tau; \phi) + 2\eta^2 F^2(\tau; \phi), \\
\frac{\partial G}{\partial \tau}(\tau; \phi) &= \kappa \, H(\tau; \phi) + 2\lambda \, F(\tau; \phi) + \nu^2 E(\tau; \phi) H(\tau; \phi) + 2\eta^2 F(\tau; \phi) G(\tau; \phi), \\
\frac{\partial H}{\partial \tau}(\tau; \phi) &= -A_j(\phi) \, H(\tau; \phi) + \lambda \, E(\tau; \phi) + 2\nu^2 D(\tau; \phi) H(\tau; \phi) + \eta^2 E(\tau; \phi) G(\tau; \phi), \\
\frac{\partial C}{\partial \tau}(\tau; \phi) &= r \, i\phi + \lambda \, G(\tau; \phi) + \tfrac{1}{2}\nu^2 H^2(\tau; \phi) + \nu^2 D(\tau; \phi) + \tfrac{1}{2}\eta^2 G^2(\tau; \phi) + \eta^2 F(\tau; \phi)
\end{aligned} \tag{31}$$

Here $u_1 = \frac{1}{2}$, $u_2 = -\frac{1}{2}$, $b_1 = 1$, $b_2 = 0$, $A_j(\phi) = \kappa - \nu\rho \, (b_j + i\phi)$ and the initial conditions $C(0; \phi) = D(0; \phi) = E(0; \phi) = F(0; \phi) = G(0; \phi) = H(0; \phi) = 0$.

Remark 3.1 The ODE system (31) differs form that presented in Lin et al. (2024) (equation (4)), where the latter includes an additional τ as multiplicative factor of $u_j\, i\phi - \frac{1}{2}\phi^2$ in the first ode $\frac{\partial D}{\partial \tau}(\tau;\phi) = \ldots$.

After careful analysis, we concluded that τ must be dropped if the European price computed using the semi-analytical formula were to match the price using a Mone Carlo simulation. Further investigation of the derivation in Lin et al. (2024) resulted in the following analytical argument in favor of dropping the τ term. Lin et al. (2024) derive the ODE system for f_j by assuming the form (30) then plugging it to the second order diffusion PDE (32) (matching equation (13) in Lin et al. (2024)):

$$\begin{aligned}\frac{1}{2}\sigma_t\frac{\partial^2 f_j}{\partial x_t^2} + \frac{1}{2}\nu^2\frac{\partial^2 f_j}{\partial \sigma_t^2} + \frac{1}{2}\eta^2\frac{\partial^2 f_j}{\partial \theta_t^2} + \big(k(\theta_t - \sigma_t) + b_j\big)\frac{\partial f_j}{\partial \sigma_t} \\ + \nu\sigma_t\rho\frac{\partial^2 f_j}{\partial x_t\,\partial \sigma_t} + (r + u_j\sigma_t^2)\frac{\partial f_j}{\partial x_t} + \lambda\frac{\partial f_j}{\partial \theta_t} + \frac{\partial f_j}{\partial \tau} - r f_j = 0\end{aligned} \tag{32}$$

The PDE coefficients are τ independent. The first and second order partial derivatives of f_j (30) do not include a τ term, rather only the state variables x_t, σ_t, θ_t, the coefficients C, D, $\ldots$, and their derivatives. After plugging f_j to (32), like terms in the state variables are grouped and set to 0 leading to ODE (31). Setting coefficients of the state variables to 0 is necessary for equation (32) to hold for all t. Throughout this process no τ terms are injected into the computation. Therefore, ODE (31) must be autonomous.

Further evidence in favor of dropping τ is that the prices obtained by solving (31) not only match the Monte Carlo price, but also prices computed independently under Black-Scholes and Stein-Stein models (Table 1, Stein and Stein (1991)), as well as the Schöbel-Zhu model (Table 2, Schöbel and Zhu (1999)).

3.1.3 Algorithm for LLH Price

For given values of x_t, σ_t, θ_t, K and τ, the LLH price of a call is computed using formula (27). The computation starts with the numerical integration of ODE (31) on a $(\tau,\phi)-$grid to estimate the values of $D(\tau,\phi)$, $E(\tau,\phi)$, $F(\tau,\phi)$, $G(\tau,\phi)$, $H(\tau,\phi)$, and $C(\tau,\phi)$. This is followed by the estimation of the integrals (29) and finally the price U. The grid is dictated by the upper limit $\phi_{\max}$ set on ϕ in the estimation of the integrals P_j, the integration step size $\Delta\phi$, and the ODE discretization step size Δt over $[t, T]$. Set the number of integration steps over $[0, \phi_{\max}]$ to N_{P_j} and the number of time steps over $[t, T]$ to N_τ. Write the ODE system (31) as $\frac{dX}{d\tau}(\tau,\phi) = f_\Theta(\tau,\phi,X)$ where

$$X(\tau,\phi) = [D(\tau,\phi),\ E(\tau,\phi),\ F(\tau,\phi),\ G(\tau,\phi),\ H(\tau,\phi),\ C(\tau,\phi)]^\top$$

and $\Theta = (r,\kappa,\nu,\lambda,\eta,\rho)$.

We implement the following algorithm for computing the price.

1. For $k = 0, 1, \ldots, N_{P_j}$, for $i = 0, 1, \ldots, N_\tau$: given x_{t_i}, σ_{t_i}, θ_{t_i}, K and $\tau_i = T - t_i$. integrate (31) by applying an order 4 Runge-Kutta scheme with step size $\Delta t = T/N_\tau$

2. For $j = 1, 2$, for $i = 0, 1, \ldots, N_\tau$, for $k = 0, 1, \ldots, N_{P_j}$ compute f_j using (30) and the values of $X(\tau_i, \phi_k)$ estimated in step 1.

3. For $j = 1, 2$ compute P_j (29) by applying the trapezoid quadrature with step size $\Delta\phi = \phi_{\max}/N_{P_j}$.

4. Compute the price using (27).

3.2 Eurpoean Put Price

European put prices under LLH dynamics are obtained via the put-call parity equation:

$$P = C - S + K\, e^{-r\tau}. \tag{33}$$

where C is the LLH price obtained from formula (27) for strike K and maturity τ.

3.3 American Call Price

The LLH model assumes no dividends which implies that the optimal exercise time is at maturity. The American and European calls have the same price.

3.4 American Put Price

An estimator of the American price is obtained following the LSM algorithm introduced in Longstaff and Schwartz (2001). We adopt the notation from Rasmussen (2002) and Aquino et al. (2022).

3.4.1 American Option Pricing Problem

The problem of pricing an American option with maturity T consists of finding an optimal exercise strategy and valuing the expected discounted payoff from this strategy. Given the underlying asset value S_t, let V_t denote the option value at time t, defined as

$$\beta_t\, V_t(S_t) = \sup_{\tau \in \mathcal{T}(t,T)} \mathbb{E}\big[\beta_\tau\, X_\tau(S_\tau) \mid S_t\big], \tag{34}$$

where $\{X_t\}_{0 \le t \le T}$ is the payoff process with discount factors $\beta_t = e^{-rt}$, $\beta_0 = 1$, and $\mathcal{T}(t,T)$ denotes the class of stopping times satisfying $t \le \tau \le T$. We assume the discount factors to be deterministic and exogenous, and the asset value process to be Markovian. For a put option the exercise price at time t is $X_t(S_t) = (K - S_t)^+ := \max\{K - S_t,\, 0\}$.

Solving (34) is simplified by restricting the set of possible exercise times, resulting in the Bermudan approximation. Allowing the option holder to exercise at equidistant times $t_j = j\Delta t$, $j = 0, \ldots, M$, with $\Delta t = T/M$, the Bermudan option value is

$$V_0^B(S_0) = \sup_{\tau \in \mathcal{T}_B(0,T)} \mathbb{E}\big[\beta_\tau\, X_\tau(S_\tau) \mid S_0\big] \;\le\; V_0(S_0), \tag{35}$$

where $\mathcal{T}_B(0,T)$ is the class of stopping times taking values in $\{0, t_1, \ldots, t_M\}$, where $t_j = j\Delta t$ for $j = 1, \ldots,\, M$ and $\Delta t = T/M$. Under mild conditions, $V_0^B(S_0)$ converges to $V_0(S_0)$ as M grows to infinity, Aquino et al. (2022). If the option has not yet been exercised at t_j, the option value can be shown to satisfy the Bellman equation

$$V_j(S_{t_j}) = \max\big\{X_{t_j}(S_{t_j}),\; \beta_{\Delta t}\, \mathbb{E}\big[V_{j+1}(S_{t_{j+1}}) \mid S_{t_j}\big]\big\}\,. \tag{36}$$

The rationale is to exercise if the exercise price surpasses the continuation value. However, the continuation value has no analytical expression and must be approximated.

3.4.2 LSM Algorithm

The key idea is that at each exercise date, the continuation value $\beta_{\Delta t}\, \mathbb{E}\left[V_{j+1}(S_{t_{j+1}})\right]$ is estimated by regressing discounted future cashflows on basis functions evaluated at the current spot prices.

More specifically, for each simulated terminal spot S_T^i, set $\hat{V}_T^i(S_T^i) = (K - S_T^i)^+$. For $j = M-1, \ldots, 1$ and paths $i = 1, \ldots, N$, the estimated option value is

$$\hat{V}_j^i(S_{t_j}^i) = \begin{cases} X_{t_j}(S_{t_j}^i) & \text{if } X_{t_j}(S_{t_j}^i) \geq \tilde{V}_j(S_{t_j}^i), \\ \beta_{\Delta t}\, \hat{V}_{j+1}^i(S_{t_{j+1}}^i) & \text{otherwise,} \end{cases} \tag{37}$$

where $\tilde{V}_j(S_{t_j})$ is the estimated continuation value at t_j. The option value estimator is

$$\hat{V}_0(S_0) = \max\left\{ X_0(S_0),\ \beta_{\Delta t}\, \frac{1}{N} \sum_{i=1}^{N} \hat{V}_1^i \right\}. \tag{38}$$

Following Longstaff and Schwartz (2001), the price (38) is computed with the regression estimate (39) of the continuation value on basis functions $\{B_k : 0 \leq k \leq J\}$, where an infinite basis is truncated to a finite set of cardinality $J+1$:

$$\beta_{\Delta t}\, \mathbb{E}\big[V_{j+1}(S_{t_{j+1}}) \mid S_{t_j}\big] \ \approx\ \tilde{V}_j(S_{t_j}) := \sum_{k=0}^{J} \alpha_j^k\, B_j^k(S_{t_j}), \tag{39}$$

Longstaff and Schwartz (2001) recommend using Laguerre polynomials as basis. We compare two scenarios where the B_j^k's are defined using Laguerre polynomials and Gaussian radial basis functions, see section 4.3. The coefficients α_j^k are estimated by ridge regression restricted to the in-the-money (ITM) set $\mathcal{N}_j = \{i : X_{t_j}(S_{t_j}^i) > 0\}$:

$$\alpha_j = \arg\min_{\alpha}\ \frac{1}{|\mathcal{N}_j|} \sum_{i \in \mathcal{N}_j} \left(\sum_{k=0}^{J} \alpha_j^k\, B_j^k(S_{t_j}^i) - \beta_{\Delta t}\, \hat{V}_{t_{j+1}}^i(S_{t_{j+1}}^i) \right)^2 + \lambda\, \|\alpha_j\|^2. \tag{40}$$

Restricting the regression to $\mathcal{N}_j$ limits the region over which the conditional expectation must be estimated, so that fewer basis functions are needed for an accurate approximation.

Remark 3.2 Following West (2013), the European price is set as an exercise barrier to prevent spurious early exercise in (37), meaning a path exercises only if:

$$X_{t_j}(S_{t_j}^i) > \max(E_j^i, \tilde{V}_j(S_{t_j}^i)) \tag{41}$$

where E_j^i is the European option value at the node.

Remark 3.3 Longstaff and Schwartz (2001) show that the LSM estimator converges to a lower bound of the true option value as $N \to \infty$; the bound tightens as $J \to \infty$. Under fairly general conditions, Clément et al. (2002) prove the almost sure convergence of the LSM algorithm as N, $J \to \infty$ and prove that its normalized error to be asymptotically Gaussian.

3.4.3 Variance Reduction Approach

The principle of control variate variance reduction in the current context is presented in West (2013) as follows. For a given Monte Carlo simulation of the paths, if A_{MC} is an estimator of the American put price and E_{MC} is an estimator of the European price, the variance of A_{MC} may be reduced via (42):

$$\hat{V} := A_{MC} - \theta(E_{MC} - E_{CF}), \tag{42}$$

where E_{CF} is the European price estimated using a closed formula with $\mathbb{E}\left[E_{MC}\right] = E_{CF}$.

E_{MC} is said to be the control variate and the resulting variance reduction under E_{MC} is optimal for

$$\theta = \frac{\mathbb{C}ov(A_{MC}, E_{MC})}{\mathbb{V}ar(E_{MC})}. \tag{43}$$

For a more general treatment, see section 4.1 of Glasserman (2004). Notice that θ is the linear regression coefficient of A_{MC} on E_{MC}. The variance reduction factor that results from introducing the control variate is obtained from (42) and (43):

$$\mathbb{V}ar\left(\hat{V}\right) = (1-\rho^2)\mathbb{V}ar\left(A_{MC}\right) \tag{44}$$

with $\rho^2 = \dfrac{\mathbb{C}ov^2(A_{MC}, E_{MC})}{\mathbb{V}ar(A_{MC})\mathbb{V}ar(E_{MC})}$. From (44), it follows that the optimal control variate is one that is most correlated with A_{MC}. This observation pins the idea behind Rasmussen's control variate.

3.4.4 Rasmussen Control Variate

Let $\bar{\tau}^i_j$ denote the first time t_k, $k \geq j$, the option is exercised along path i. Rasmussen (2005) suggests to control the LSM estimated price $\hat{V}^i_j$ with $e^{-r(\bar{\tau}_{j+1}-t_j)}E^i_{\bar{\tau}^i_{j+1}}$ whose expectation is given by the optimal sampling theorem:

$$\mathbb{E}_{t_j}\left[e^{-r(\bar{\tau}^i_{j+1}-t_j)}E^i_{\bar{\tau}_{j+1}}\right] = E^i_{t_j},$$

where $E^i_{t_j}$ is the European option value at node $S^i_{t_j}$ with maturity $T - t_j$, computed via a closed-form formula, West (2013); The control parameter follows from (43):

$$\theta^i_j = \frac{\mathbb{C}ov\left(e^{-r(\bar{\tau}^i_{j+1}-t_j)}I^i_{\bar{\tau}^i_{j+1}},\, e^{-r(\bar{\tau}^i_{j+1}-t_j)}E^i_{\bar{\tau}^i_{j+1}}\right)}{\mathbb{V}ar\left(e^{-r(\bar{\tau}^i_{j+1}-t_j)}E^i_{\bar{\tau}_{j+1}}\right)}. \tag{45}$$

West (2013, section 3.1) demonstrates that this control variate maximizes correlation with the American price and suggests to estimate the control coefficients $\hat{\theta}^i_j$ as follows:

$$\hat{\theta}^i_j = \frac{\widehat{CV}^{\text{aux}_1}_j(S^i_{t_j}) - \tilde{V}_j(S^i_{t_j})\,\tilde{V}^E_j(S^i_{t_j})}{\widehat{CV}^{\text{aux}_2}_j(S^i_{t_j}) - \left[\tilde{V}^E_j(S^i_{t_j})\right]^2} \tag{46}$$

where:

(i) $CF^i_j = e^{-r(\bar{\tau}^i_{j+1}-t_j)}I^i_{\bar{\tau}^i_{j+1}}$, the discounted eventual exercise value (or payoff $X_{\bar{\tau}^i_{j+1}}(S_{\bar{\tau}^i_{j+1}})$) along path i, where $\bar{\tau}^i_{j+1}$ is the exercise time and $I^i_{\bar{\tau}^i_{j+1}}$ is the intrinsic value at exercise;

(ii) $\tilde{V}^E_j(S^i_{t_j})$: the modeled European value, obtained by regressing $CV^i_j = e^{-r(\bar{\tau}^i_{j+1}-t_j)}E^i_{\bar{\tau}^i_{j+1}}$ on the same basis; CV^i_j the European value tracked along the exercise boundary (48);

(iii) $\widehat{CV}^{\text{aux}_1}_j(S^i_{t_j})$ and $\widehat{CV}^{\text{aux}_2}_j(S^i_{t_j})$: auxiliary estimators from regressing $CF^i_j \cdot CV^i_j$ and $(CV^i_j)^2$ on the same basis.

Remark 3.4 Following the introduction of Rasmussen's control variate, the computation of the LSM estimator of the American price is altered by:

(i) modifying the test value $\tilde{V}_j(S^i_{t_j})$ used for the exercise decision in (37) to TV^i_j:

$$TV^i_j = \tilde{V}_j(S^i_{t_j}) - \hat{\theta}^i_j\big(\tilde{V}^E_j(S^i_{t_j}) - E^i_j\big) \tag{47}$$

(ii) resetting the exercise boundary (41) to (48):

$$X_{t_j}(S^i_{t_j}) > \max(E^i_j,\ TV^i_j) \tag{48}$$

3.4.5 LSM+CV-LLH Algorithm

1. At maturity t_N initialize CF and CV as the exercise price (or payoff) of the option, based on the simulated paths of S.

2. For $j = N-1, \ldots, 1$:

   (i) Apply the updates $CF_j = e^{-r\Delta t_{j+1}} CF_{j+1}$ and $CV_j = e^{-r\Delta t_{j+1}} CV_{j+1}$

   (ii) Compute the estimators: $\tilde{V}_j(S_{t_j})$, $\tilde{V}^E_j(S_{t_j})$ by regressing CF_j and CV_j respectively on the given basis functions $B_k(S^i_{t_j}/K)$. The regressions are fitted over in-the-money paths only ($I^i_j > 0$) following Longstaff and Schwartz (2001).

   (iii) Compute the auxiliary estimators $\widehat{CV}_j^{\text{aux}_1}(S_{t_j})$ and $\widehat{CV}_j^{\text{aux}_2}(S_{t_j})$ by regressing the values $E^i_j CV^i_j$, and the values $(CV^i_j)^2$ on the same set of basis functions.

   (iv) Compute the control parameters $\hat{\theta}^i_j$ (46)

   (v) Compute the test value TV (47)

3. If the immediate exercise value $I^i_j > \max(E^i_j, TV^i_j)$, then apply the updates:

$$CF^i_j = I^i \text{ and } CV^i_j = E^i_j$$

4. The option value $\hat{V}_{\text{base}}$ determined by the LSM combined with the Rasmussen control variate is

$$\hat{V}_{\text{base}} = \max\left(I_0,\ \frac{e^{-rt_1}}{N}\sum_{i=1}^{N} CF^i_1\right)$$

5. An improved estimator $\hat{V}_{\text{improved}}$ is reported in West (2013) following the example of Rasmussen (2005):

$$\hat{V}_{\text{improved}} = \max\left(I_0,\ \frac{e^{-rt_1}}{N}\sum_{i=1}^{N} CF^i_1 - \hat{\theta}\left(\frac{e^{-rt_1}}{N}\sum_{i=1}^{N} CV^i_1 - E^i_1\right)\right) \tag{49}$$

   where $\hat{\theta}$ is the regression coefficient of CF_1 on CV_1 give by:

$$\hat{\theta} = \frac{N\sum_{i=1}^{N} CF^i_1 CV^i_1 - \sum_{i=1}^{N} CF^i_1 \sum_{i=1}^{N} CV^i_1}{N\sum_{i=1}^{N}(CV^i_1)^2 - \left(\sum_{i=1}^{N} CV^i_1\right)^2} \tag{50}$$

3.4.6 Computational Complexity

The computational cost of computing an LSM estimator of the American price with Rasmussen control variate under LLH dynamics (LSM+CV-LLH) can be tracked by accounting for the cost of path simulation, the backward loop operations including the recursion (37)-(38) and regression (40), as well as the operations underlying the control variate computation including (29)-(31) and (46)-(50).

Denote by N the count of simulated MC paths, P the number of quadrature nodes used in the computation of characteristic functions, and N_{RK} the number of Runge-Kutta integration steps for the ODE system. Table 2 decomposes the per-step and total cost of the LSM+CV-LLH algorithm. With $N \gg P \gg J$, the dominant term is the European pricing in the backward loop:

$$\text{Cost(LSM+CV-LLH)} = O(MNP).$$

Figure 2 verifies the underlying linear scaling in each parameter.

Stage	Per-step cost	Total cost
Path simulation	$O(N)$	$O(MN)$
ODE precomputation (Ode + f_j's)	$O(PN_{RK} + N)$	$O(MN)$
European pricing E_j (Quadrature)	$O(PN)$	$O(MNP)$
Regression ($\Phi^\top\Phi$ + solve + predict)	$O(NJ^2 + NJ)$	$O(MNJ^2)$
Exercise / CV updates	$O(N)$	$O(MN)$

Table 2: Computational complexity of LSM+CV-LLH.

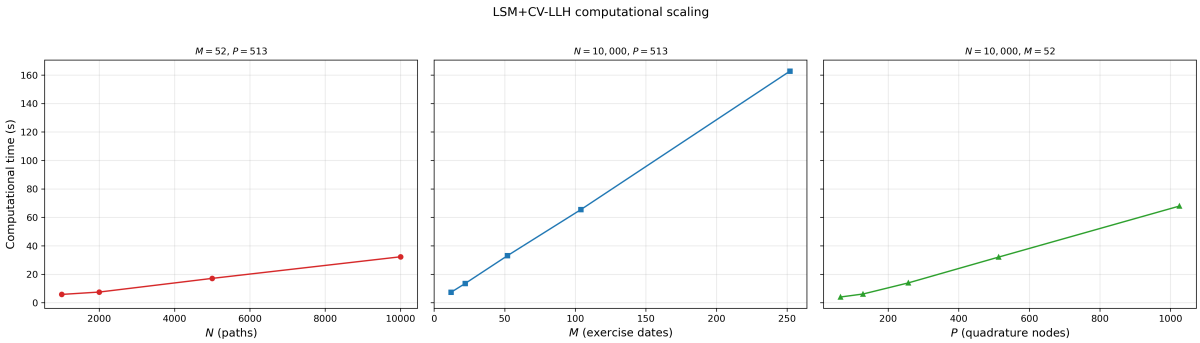


Figure 2: Time cost scaling with N paths, M exercise dates, and P quadrature nodes.

4 Experiments

To assess robustness across calibration regimes, all experiments are repeated under the two reference parameter sets provided in Table 1. The numerical parameters used in producing the results are summarized in Table 3. Three exercise floor methods are compared, where the European price E_j^i in (48) is computed using the LLH formula (LLH price), the Black-Scholes formula (BS price), or as the time t_j discounted payoff of the option along path S_t^i (MC1 price).

4.1 Experimental Setup

All experiments were run on an Intel Core i7-13700HX (16 cores, 16 GB RAM) using Python 3.12 (NumPy 1.26.4, SciPy 1.13.1) with double-precision arithmetic. Random variates are drawn via NumPy's PCG64 generator (Permuted Congruential Generator, 128-bit state) with fixed seeds for reproducibility.

Table 3: Numerical control parameters.

Computation	Parameters	Value
Simulation (1-month & 1-year)	N_{paths}	up to 50,000
	n_{steps}	22/ 52
	seed	42
LSM regression (2 bases)	basis type	Laguerre/ Gaussian RBF
	basis order	up to deg 2/ 10 RBFs
	ridge	10^{-5}/ 10^{-4}
Exercise floor	floor method	LLH price
		BS price
		MC1 price
European pricing (ODE & quadrature)	$\phi_{\max}$	300
	n_{ϕ}	513
	$n_{\text{steps, ODE}}$	256

4.2 European Pricing Experiments

A semi-analytical European pricing formula is derived in Lin et al. (2024) via characteristic functions, computed by solving numerically a system of ODEs followed by numerical Fourier inversion. Following their example, we test the validity of our implementation of the formula by comparing it against Monte Carlo estimates across moneyness and maturity scenarios; testing against edge-case recovery of Black–Scholes, Stein–Stein, and Schöbel–Zhu prices; studying the sensitivity of the price to the LLH extension parameters λ and η. A Jupyter notebook that illustrates the workflow can be found at this link.

4.2.1 Experiment 1: LLH Formula vs Monte Carlo Pricing

We compare the LLH formula and Monte Carlo prices across the moneyness ladder as shown in Figure 3. At strike $K = 100$, the spot grid ranges from deep out-the-money (OTM) to deep in-the-money (ITM) values. The time to maturity is $\tau = 1$ year and the number of MC paths is 10^6 with 52 steps. We observe that for LLH Table 1 regime, the formula and MC prices agree closely to within 3%, while for LLH Table 2 regime, the relative error is within 1%. The difference in the error bounds suggests that at higher drift parameter values, smaller time steps may be required in the discretization and integration methods or a higher count of simulated MC paths.

4.2.2 Experiment 2: Computational Performance

Table 4 compares the computational cost of the LLH semi-analytical formula (ODE integration and trapezoid quadrature) against a Monte Carlo simulation with 5×10^5 paths. Timings are the median of five runs under LLH Table 1 regime with $S_0 = 100$, $K = 90$. The semi-analytical formula is significantly faster than Monte Carlo, with the speedup growing for longer maturities with increased MC time steps. This suggests the LLH formula as the preferred method for European pricing under LLH dynamics.

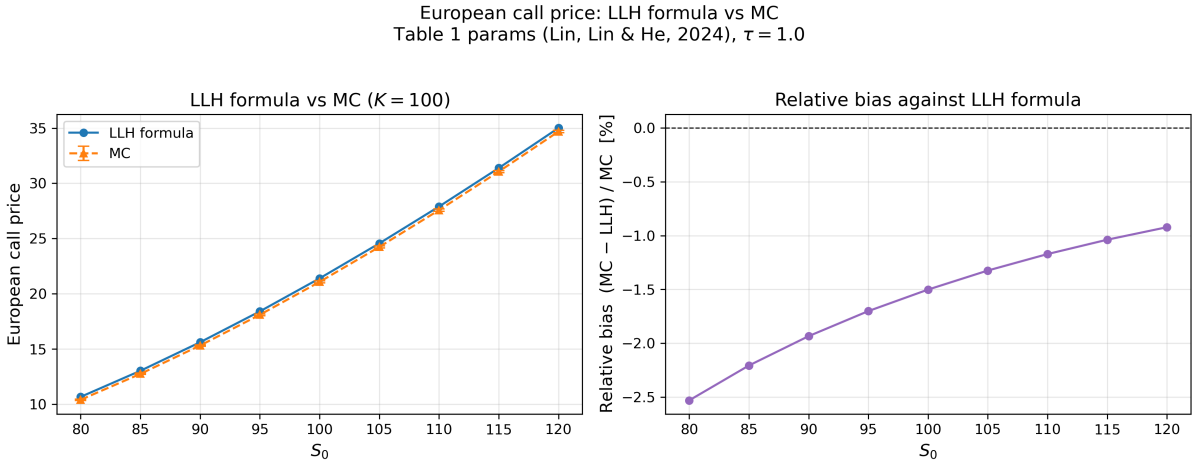


Figure 3: LLH formula vs MC — LLH Table 1 regime, $\tau = 1.0$, 10^6 paths.

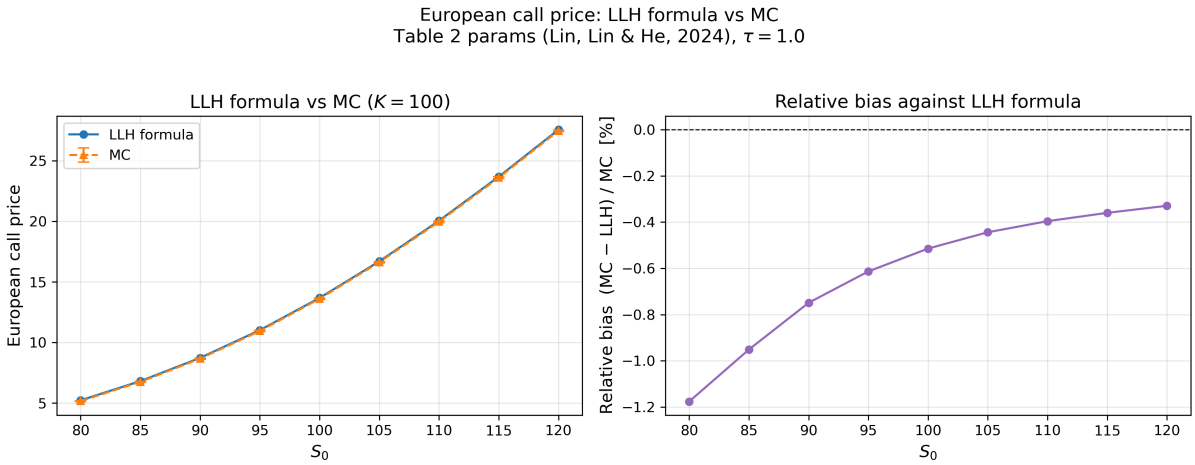


Figure 4: LLH formula vs MC — LLH Table 2 regime, $\tau = 1.0$, 10^6 paths.

Table 4: Computational cost of LLH formula vs Monte Carlo.

Horizon	LLH (s)	MC (s)	Speedup
1 month ($\tau = 1/12$, 22 steps)	$\sim$0.02	$\sim$0.5	$\sim$25$\times$
1 year ($\tau = 1.0$, 52 steps)	$\sim$0.02	$\sim$1.2	$\sim$60$\times$

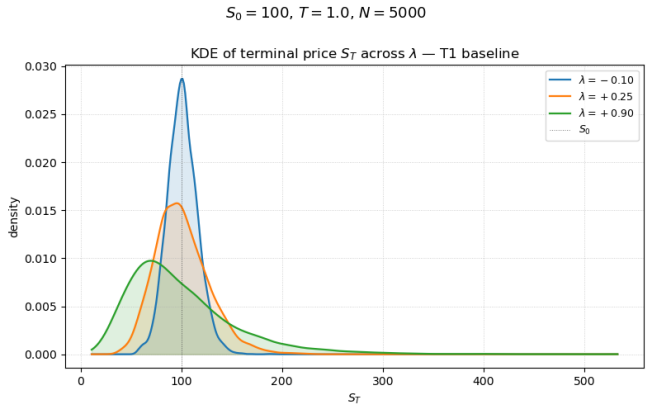


Figure 5: λ sensitivity of S_T distribution, Table 1 regime

4.2.3 Experiment 3: Edge-Case Validation

The LLH model nests three classical models as special cases. We verify that the implementation recovers the known prices in each limit, see section 5 in the notebook at this link.

Black–Scholes Limit ($\kappa = \nu = \lambda = \eta = \rho = 0$). In this limit:

- The LLH formula price matches the Black–Scholes closed form to machine precision.
- Simulated terminal prices pass the Kolmogorov–Smirnov test for lognormality, confirming correct path generation.

Parameters: $r = 0.05$, $\sigma_0 = 0.2$, $\theta_0 = 0$, $S_0 = 100$, $K = 90$, $\tau = 1.0$.

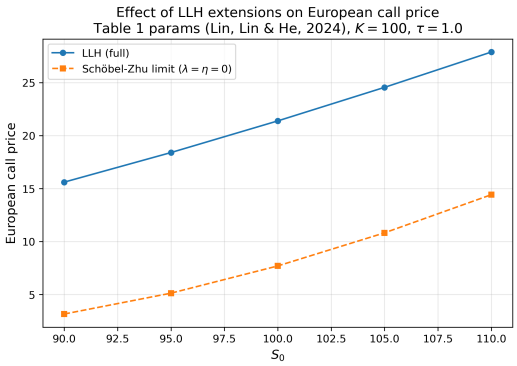
Stein–Stein Limit ($\lambda = \eta = \rho = 0$). The LLH formula returns a call price consistent with Stein and Stein (1991), Table 1 reference value of 10.77.
Parameters: $r = 0.0953$, $\kappa = 4.0$, $\nu = 0.1$, $\sigma_0 = 0.2$, $\theta_0 = 0.2$, $S_0 = 100$, $K = 100$, $\tau = 1/12$.

Schöbel–Zhu Limit ($\lambda = \eta = 0$). In all tested configurations, the implemented LLH formula recovers to four decimal places Schöbel and Zhu (1999), Table 2 displaying European prices across $\rho \in \{0.5, 0.0, -0.5\}$, $\theta_0 \in \{0.0, 0.1, 0.2, 0.3\}$, and $K \in \{90, 95, 100, 105, 110, 115, 120\}$.
Parameters: $S_0 = 100$, $\tau = 0.5$, $r = 0.0953$, $\kappa = 4$, $\nu = 0.1$, $\sigma_0 = 0.15$.

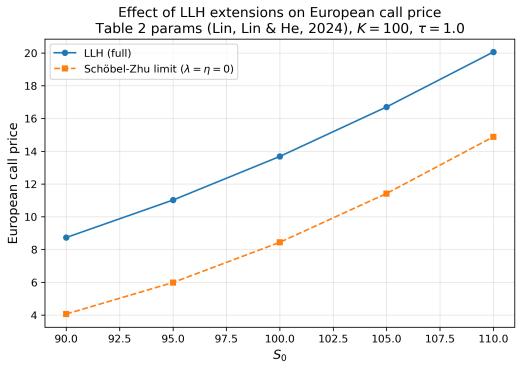
4.2.4 Experiment 4: Effect of LLH Extensions over the Schöbel–Zhu Limit

We test our implementation against properties expected of the pricing formula under the LLH dynamics. The formula must price the impact of λ and η on volatility in a consistent fashion. For example, under LLH Table 1 regime, the drift parameter $\lambda = 0.9$ substantially increases option prices relative to SZ price across all spots, reflecting a rising long-run mean that pushes the future volatility distribution upward, see Figures 5 and 6. Under LLH Table 2 regime, the impact is smaller, consistent with the moderate $\lambda = 0.3115$. Note that the Schöbel–Zhu price is computed using the LLH formula with $\lambda = \eta = 0$.

To further test the LLH price sensitivity to λ relative to Schöbel–Zhu's, we sweep $\lambda \in [-0.2, 0.5]$ while holding all other parameters fixed, Figure 7. As expected, the formula harvests less premium with decreasing λ since the volatility reverts moderately to a smaller mean. Notice how larger η in LLH Table 2 regime exacerbates the price increase over the SZ benchmark.

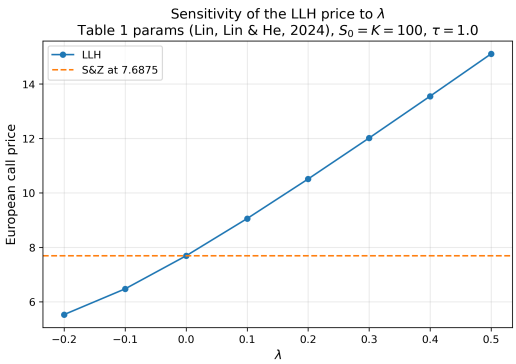


(a) LLh Table 1 regime

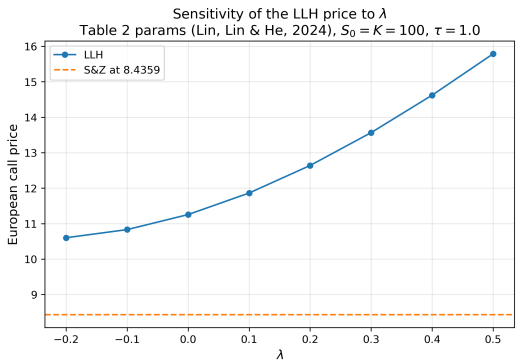


(b) LLH Table 2 regime

Figure 6: LLH vs. SZ price— evaluated with common parameters.

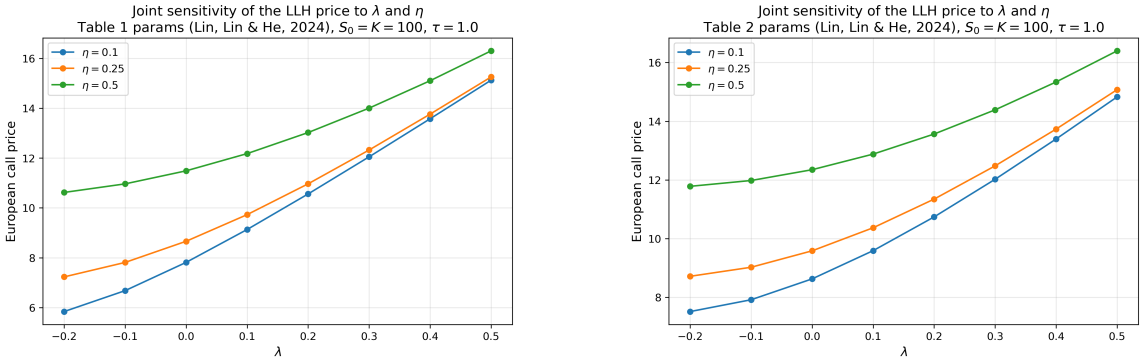


(a) LLh Table 1 regime



(b) LLh Table 2 regime

Figure 7: λ sensitivity of LLH price



(a) LLH Table 1 regime

(b) LLH Table 2 regime

Figure 8: $\lambda - \eta$ joint sensitivity of LLH price — $\tau = 1.0$.

4.2.5 Experiment 5: Joint Sensitivity to λ and η

The parameters λ and η are the two additions of the LLH model relative to the Schöbel–Zhu framework. We overlay curves for $\eta \in \{0.1, 0.25, 0.5\}$, each sweeping $\lambda \in [-0.2, .5]$, Figure 8. As expected, higher η amplifies the price sensitivity to λ. The underlying reason is that η controls the diffusion of θ_t, and larger diffusion widens the distribution of future volatility levels.

4.3 American Price Experiments

The central question is whether Rasmussen control variates remain effective under stochastic volatility as they are under geometric Brownian motion, Aquino et al. (2022). Given the semi-analytical nature of the LLH formula, another question is the computational stability and cost of using a Rasmussen-style control variate. We investigate these questions by comparing across moneyness and parameter regimes variants of two estimators: plain LSM (38) and LSM combined with Rasmussen control variate (49), hereafter referred to as LSM+CV-LLH. The variants differ by the computation method of the exercise floor (41). Variance reduction and early exercise premium (EEP) are computed and reported.

All experiments are repeated under the LLH parameter sets in Table 1 to assess robustness across calibration regimes. The numerical parameters used in producing the results are summarized in Table 3. At each time step, the basis functions (39) are defined using Laguerre polynomials or RBF regression centered at equally-spaced quantiles of the ITM moneyness values $\{S^i_{t_j}/K : i \in \mathcal{N}_j\}$. For RBF regression, the shared bandwidth h is set to the median spacing between consecutive centers. Both the centers and the bandwidth are recomputed at every exercise date. European prices are precomputed and cached by maturity to avoid redundant solves across paths.

4.3.1 Experiment 1: Price vs Spot

The price is estimated at strike $K = 100$, on a spot grid $S_0 \in \{85, 90, 100, 110, 115\}$ (deep ITM to deep OTM for puts) and 1-year horizon ($T = 1.0$). Four estimators are computed using $N = 10,000$ MC paths: plain LSM and LSM+CV-LLH, each with regression bases Laguerre polynomials and Gaussian RBFs, see setup in Table 3 and results in Figure 9. The LLH European price is used as an exercise floor for all estimators. Error bars represent 95% confidence intervals.

It is observed under LLH Table 1 parameters, LSM+CV-LLH estimate is closer to the true value of the option than plain LSM in agreement with remark 3.3. The estimator harvests positive early exercise premium (EEP) across spot levels.

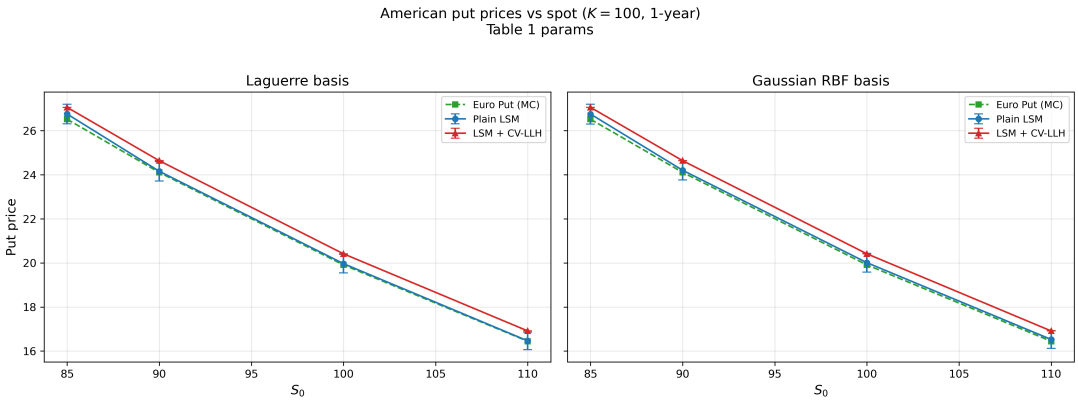


Figure 9: American put prices vs spot ($K = 100$) — Table 1 regime, $\tau = 1.0$.

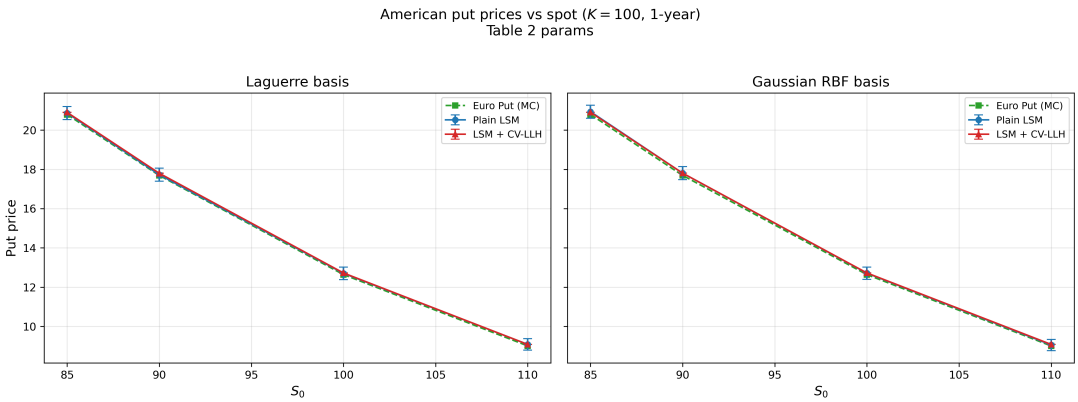


Figure 10: American put prices vs spot ($K = 100$) — Table 2 regime, $\tau = 1.0$.

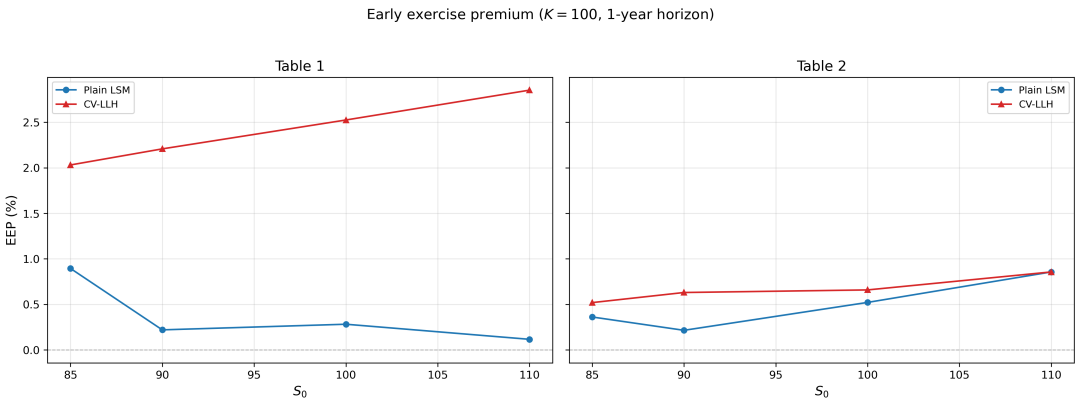


Figure 11: American put early exercise premium, $\tau = 1.0$.

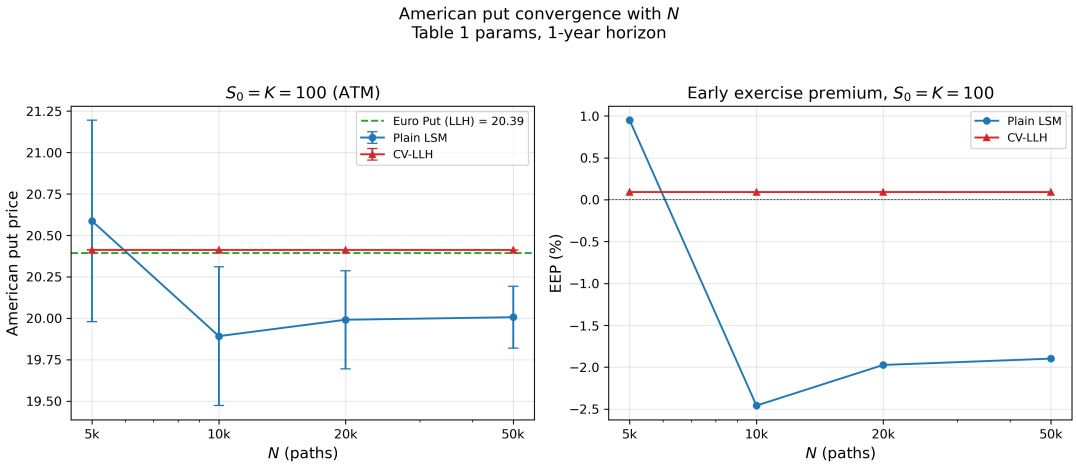


Figure 12: American price and EEP convergence with path count N — Table 1 regime.

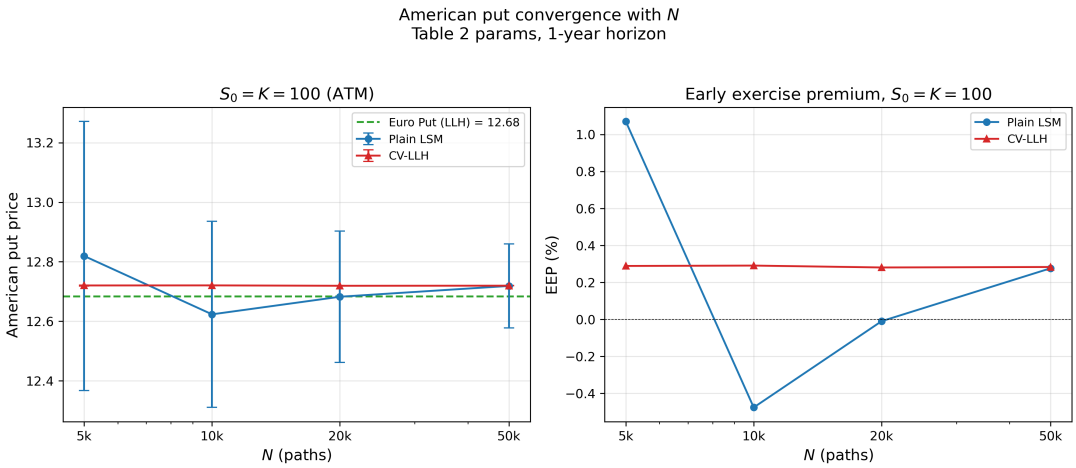


Figure 13: American price and EEP convergence with path count N — Table 2 regime.

It has been noted in section 2.3, LLH Table 2 parameters yield negative volatility. Nevertheless, LSM prices remain consistent under this regime, see Figures 10 and 11.

4.3.2 Experiment 2: Convergence with Path Count

The Plain LSM and LSM+CV-LLH estimators are computed under regression on Laguerre basis for an at-the-money (ATM) put ($S_0 = 100$, $K = 100$). Figures 12 and 13 show that LSM+CV-LLH estimator converges fast with increasing number of simulation paths N, in agreement with the results in Rasmussen (2005).

4.3.3 Experiment 3: Black–Scholes Limit Validation

Setting the LLH parameters to ($\kappa = \nu = \lambda = \eta = \rho = 0$), $r = 0.05$ and $\sigma_0 = 0.2$ results in a geometric Brownian motion model of S_t. The LLH formula and exercise floor reduce to the BS price. With these parameters and European pricing method, we compare the plain LSM and LSM+CV-BS estimators, Figure 14. LSM+CV-BS matches plain LSM in agreement with remark 3.3, both at positive EEP. The variance reduction ratios of LSM+CV-BS shown in Figure 14 are in agreement with the results shown in Table 1.3 of Aquino et al. (2022). Furthermore, it is observed that plain LSM is much more stable under GBM, suggesting the need for higher MC path count under a full LLH model, high enough to capture fat tails the model might

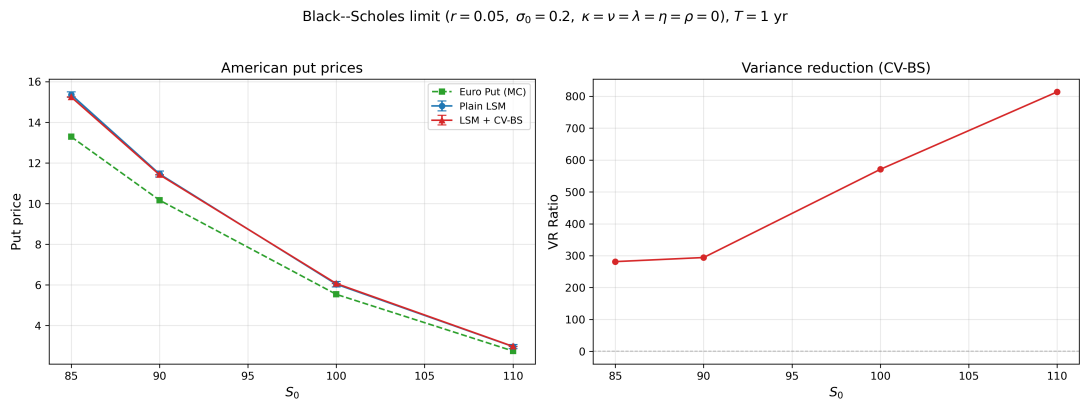


Figure 14: American put prices — Black–Scholes limit, $\tau = 1$ year.

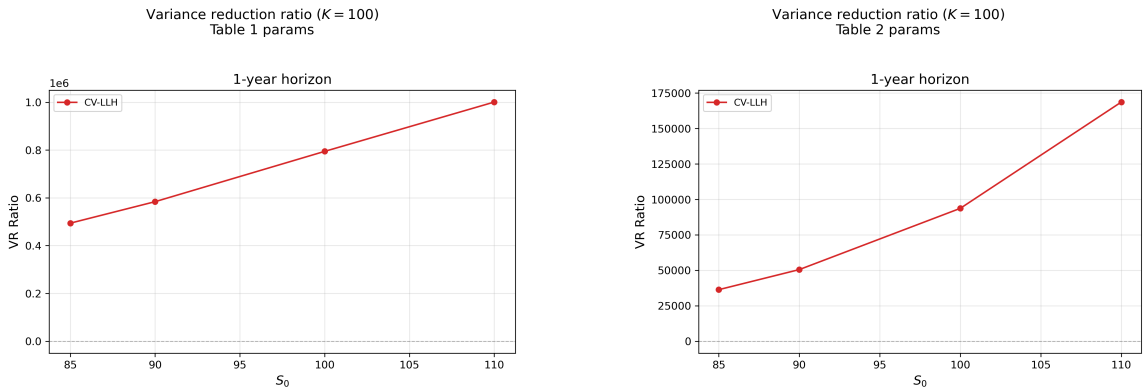


Figure 15: Variance reduction factor, 1 year maturity.

exhibit, Figure 1. Both estimators under Black-Scholes model demonstrate behavior that further validates our implementation.

4.3.4 Experiment 4: Variance Reduction Analysis

The variance reduction factor resulting from a control variate is $1/(1-\rho^2)$, as shown in section 3.4.3. Figure 15 reveals the substantial variance reduction gain when a Rasmussen control variate is combined with the LSM estimator, up to a million times smaller variance of the LSM+CV-LLH estimator as compared to plain LSM.

4.3.5 Experiment 5: Computational Cost Analysis

The analytical complexity breakdown in Table 2 suggests that the LLH European pricing dominates the LSM+CV-LLH runtime, scaling as $O(MNP)$. Figure 16 confirms this empirically at default parameters Table 3, where European LLH pricing accounts for the vast majority of computation time at 264 times longer runtime for LSM+CV-LLH than plain LSM.

4.3.6 Experiment 6: Exercise Floor Comparison

The exercise floor (48) prevents spurious early exercise by requiring that the intrinsic value exceed both the regression continuation estimate and a European option value. As the LLH European floor largely extends the computation time of the Plain LSM estimator, we compare with cheaper exercise floor alternatives. Four configurations of LSM are considered: pure LSM (no floor, as in (38), a single-path Monte Carlo floor $e^{-r(T-t_j)}(K - S_T^i)^+$ (MC1), a Black–Scholes floor evaluated at instantaneous vol (BS price), and an LLH floor (LLH price).

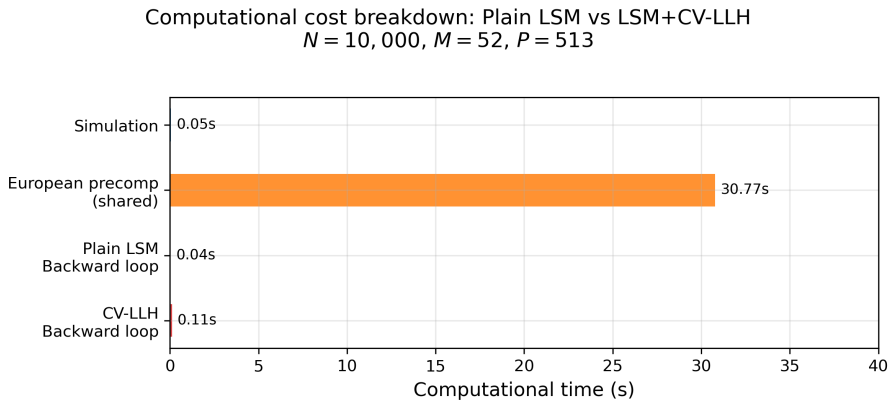
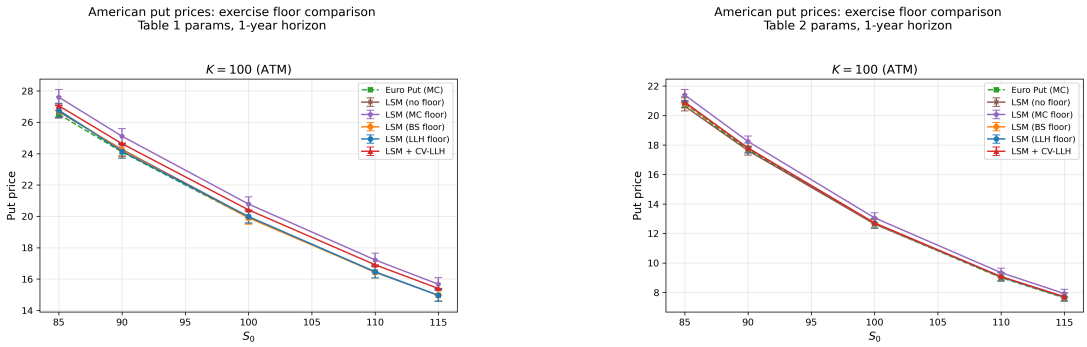


Figure 16: Computational cost breakdown — plain LSM vs LSM+CV-LLH.



(a) LLH Table 1 regime

(b) LLH Table 1 regime

Figure 17: LSM+CV-LLH exercise floor comparison.

Figure 17 compares all LSM variants alongside LSM+CV-LLH across moneyness. It is observed that under both parameter regimes in Table 1, plain LSM with MC1 price as exercise floor is a competitive baseline as compared to estimators with different exercise floor methods. It is cheap to compute and consistently dominating plain LSM with LLH floor, Figure 17).

A cost–precision tradeoff is shown in Figure 18, where all five LSM estimators evaluated at $S_0 = K = 100$ under LLH Table 1 regime.

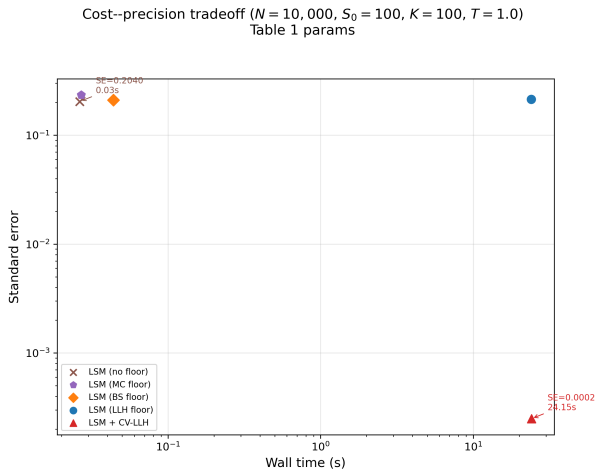


Figure 18: Cost–precision tradeoff — Table 2 regime.

5 Summary and Conclusion

We have tested our implementation of the LLH pricing formula against the prices of edge case models (BS, SS, SZ) evaluated independently under different sets of parameters. The results show agreement to within four decimal digits consistently across models. The implemented LLH price has shown accurate to within 3% of the MC price across moneyness and parameter regimes. A sensitivity analysis revealed that the implemented LLH formula is consistent in capturing the impact of the λ and η parameters that augment the SZ model. Furthermore, the LLH pricing formula demonstrates a considerable gain in speed at least $\times 60$ times for 1 year maturity and 52 weekly steps, thus suggesting that the LLH formula is the recommended method for pricing under the LLH dynamics. These results strongly establish the validity of our implementation.

We have also implemented and validated an American put pricing approach under the LLH model that combines the LSM algorithm with Rasmussen control variates. The LSM+CV-LLH estimator was computed under two reference parameter regimes, using each of two regression bases, and across the full moneyness ladder. Furthermore, we compared the price estimator with a plain LSM pricer (no control variate) under varying exercise floors. Our findings can be summarized as follows:

(a)
- **Plain LSM is unstable under full LLH dynamics.** At $50,000$ paths the plain estimator remains noisy and short of convergence; Figures 12 and 13.
- **LSM+CV-LLH is stable under LLH dynamics.** It shows to converge at low MC path count ($\sim$5K), under both reference parameter regimes, across moneyness levels, no matter the regression basis, Figures 9, 10, 12, and 13.

(b)
- **LSM+CV-LLH estimate harvests positive early exercise premium across spot levels while consistently dominating plain LSM**. This is consistent with the theoretical results described in remark 3.3, where LSM+CV-LLH takes values closer to the true value of the option than plain LSM, Figures 9, 10, and 11.
- **Rasmussen control variates with the semi-analytical LLH formula are effective under stochastic volatility.** LSM+CV-LLH delivers substantial variance reduction ranging from 0.45 to 0.5 million times smaller variance than plain LSM. The result holds under both reference parameter regimes , across moneyness levels, no matter the regression basis, Figure 15;

(c) **LLH European price is computationally expensive**. The LLH price (used as exercise floor and control variate) dominates wall time by 264 times longer runtime at default settings (Figure 16), with empirical scaling matching the analytical $O(MNP)$ complexity of Section 3.4.6 (Figure 2).

(d) **Plain LSM with a single MC path exercise floor is competitive** against plain LSM under BS and LLH floors, carries no computational consistently dominating plain LSM, Figure 17. However, without control variate the LSM estimator remains noisy at $50,000$ paths, regardless of the floor method.

When the LLH model collapses to BS, LSM+CV-BS agrees with plain LSM pricing (Figure 14) both at much lower variance, thus matching results in Aquino et al. (2022) and further validating the implementation.

Conclusion. Under LLH dynamics, Rasmussen control variate forces convergence of the LSM estimator under low MC path count, delivers strong variance reduction and shows a consistently positive early exercise premium. However, it incurs significant computation time. The computational cost is at 264 longer runtime than plain LSM alternatives. When the time budget is limited, plain LSM with a single-path Monte Carlo exercise floor offers most of the qualitative

behavior at negligible cost and is a recommended baseline. LSM+CV-LLH remains preferable when a tight confidence interval on the American price is required.

6 Future Work and Improvements

(a) Estimate vanilla option Greeks under the LLH model.

(b) Validate mathematically the $\lambda - \eta$ sensitivity observations made in 4.2.5 (prove increasing price wrt to volatility parameters, establish price convexity, etc.). Investigate LLH model sensitivity wrt the other model parameters.

(c) Compare the Rasmussen framework with other variance reduction approaches, such as using antithetic variables.

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